NR2DT-08: Step 8 — Validate

> **Series:** NR2DT | **Notebook:** 8 of 10 | **Created:** April 2026 | **Last Updated:** 04/17/2026

Overview

Goal of this step: run the three-tier validation across every migrated artifact. Syntax, tenant, and behavioral checks must all pass before cutover.

Procedural — see **NRLC-08** (Validation, Diff & Rollback) for the validation theory and tooling depth.

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Prerequisites

Requirement	Details
Audience	Migration lead + assigned engineer for this step
Completed	NR2DT-07 — Migrate Logs, Tags & Drops
Format	Procedural step — use as a runbook; defer to NRLC for depth
NRLC deep dives	NRLC-08 (Validation deep dive)

1. Three-Tier Validation Pass

The three validation tiers map onto three real `migrate.py` subcommands (there is no single `validate` subcommand — each tier uses the purpose-built command).

Tier 1 — Syntax

Re-run for any changed translated queries (the `compile --validate` path runs parser-level validation with auto-fix):

```
python3 migrate.py compile --validate --file translated-queries.csv
```

Tier 2 — Tenant

Confirm referenced metrics, entities, buckets, and OpenPipeline enrichment attributes exist in the target tenant:

```
python3 migrate.py preflight
```

Common Tier 2 failures:

- Migrated alert references a metric the tenant doesn't ingest
- DQL filter on `environment` attribute that no enrichment rule produces
- Synthetic location `AWS_US_EAST_1` not enabled in this tenant

Tier 3 — Output Parity (Behavioral)

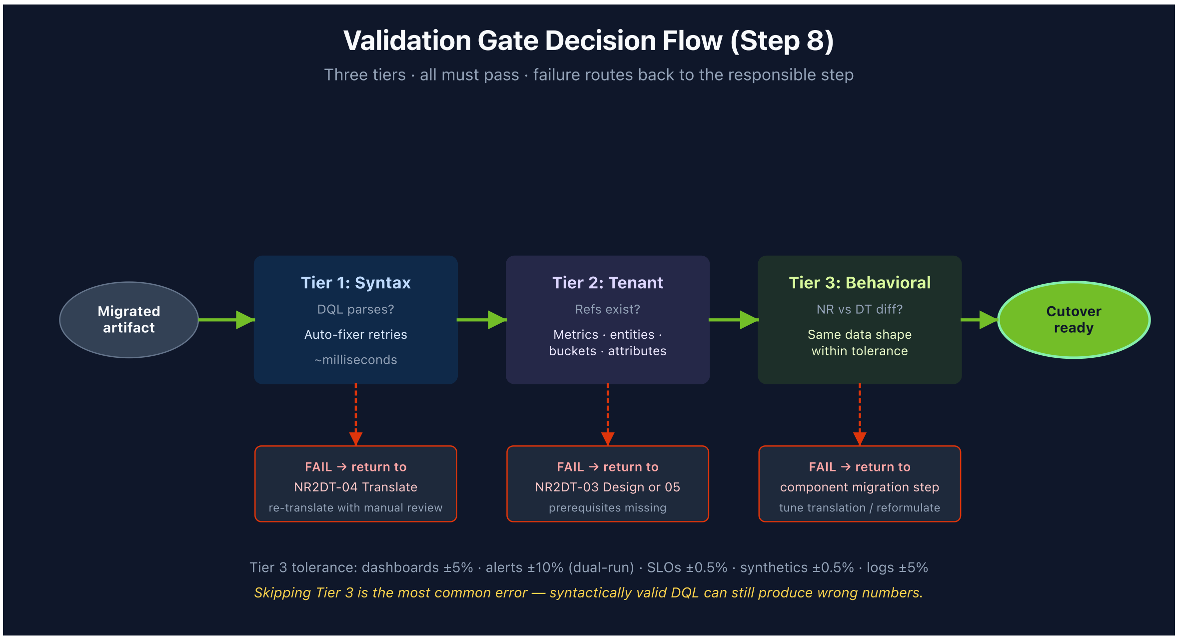
Compare NR vs. DT outputs across the dual-run window using the drift-audit command against a captured baseline:

```
python3 migrate.py audit --baseline baseline-counts.json
```

> The baseline file is produced by exporting NR + DT counts across the dual-run window; the audit command compares live DT output against those values. See NRLC-08 §4 for the capture pattern.

Tolerance per artifact:

Artifact	Tolerance
Dashboards	±5%
Alerts (dual-run volume)	±10%
SLOs	±0.5%
Synthetics	±0.5%
Log volumes	±5%



2. Diff Against Live Config

Confirm no drift since import:

```
python3 migrate.py migrate --diff --components all
```

Investigate any `EXISTING_DRIFT` entries — someone manually changed a migrated artifact in DT. Decide whether the change should be incorporated into the migration source-of-truth (Terraform), reverted, or accepted.

3. Generate the Quality Report

```
python3 migrate.py migrate --report --output validation-report.html
```

Report sections:

- Per-component pass/fail counts
- Failed translations (with reasons)
- Manual-review queue (APM conditions, scripted browsers, deferred queries)
- Rollback-eligible entities (within retention window)
- Recommended next actions

Attach this report to the cutover change-management ticket.

4. Step Exit Criteria

G8 — Cutover Ready

- [ ] Tier 1 (syntax) — all components pass
- [ ] Tier 2 (tenant) — all references resolve
- [ ] Tier 3 (behavioral / output parity) — all artifacts within tolerance
- [ ] Diff report shows no unexpected drift
- [ ] Quality report attached to cutover ticket
- [ ] Stakeholders briefed on remaining manual-review items

Next step: NR2DT-09 — Cutover, Rollback & Decommission.
